llmXive Automated Review of Scalable Visual Pretraining for Language Intelligence

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper makes several specific quantitative and citation-based claims that require verification against the provided bibliography and internal consistency.

First, there is a clear citation error in Section 2 regarding `li2026tracing`. The text uses this reference to support a mechanistic claim about SFT loss and geometric structure. However, the bibliography entry is dated 2026. Since the paper is being reviewed now, a 2026 publication cannot exist to support this claim. This suggests either a typo in the year (likely 2024 or 2025) or a hallucinated reference. This must be corrected to ensure the claim is supported by an existing source.

Second, the paper cites `qwen3.5` and `yang2025qwen3` as evaluated models. The bibliography entry for `qwen3.5` is missing the full metadata (title, journal, etc.), appearing only as a key. While `yang2025qwen3` is listed as a 2025 paper, the presence of other future-dated citations (e.g., `zhou2026scientists`, `li2026tracing`) raises a flag about the existence of these specific model versions. If these models are hypothetical or not yet public, the results presented in Table 1 and Table 2 cannot be reproduced or verified, which undermines the central empirical claim. The authors must confirm these are real, accessible models.

Third, there is a numerical inconsistency in the "Effectiveness" paragraph of Section 2. The text states: "GPQA Diamond improves by up to 3.22 points across the four backbones (e.g., 76.24 to 79.29 on Qwen 3.5)". The difference between 79.29 and 76.24 is 3.05. The value 3.22 actually corresponds to the improvement on Llama 3.1 (47.10 - 43.88 = 3.22). The text incorrectly pairs the Llama 3.1 gain with the Qwen 3.5 example. This is a factual error in reporting the results.

Finally, the claim that VP uses "only 25% of the token budget" is potentially misleading. The text clarifies that the scientific-PDF corpus yields 20B visual tokens vs 80B text tokens (a 25% ratio). However, the total CPT budget is 180B for TP and 120B for VP. The phrasing "using only 25% of the token budget" could be misinterpreted as the total training cost being 25% of the baseline, which is not the case (120B is ~67% of 180B). The claim should be qualified to specify that the 25% efficiency applies strictly to the scientific-PDF subset representation, not the total training compute.

These issues are primarily editorial and citation-related but affect the accuracy of the reported evidence. Correcting the citation year, verifying model existence, fixing the arithmetic error, and clarifying the efficiency claim will align the text with the evidence.

Action items.

- [**writing**] The paper makes several specific quantitative and citation-based claims that require verification against the provided bibliography and internal consistency. First, there is a clear citation error in Section 2 regarding li2026tracing. The text uses this reference to support a mechanistic claim about SFT loss and geometric structure. However, the bibliography entry is dated 2026. Since the paper is being reviewed now, a 2026 publication cannot exist to support this claim. This suggests either a t

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure illustrates the TP and VP pathways clearly, but the t-SNE visualization contradicts the caption's claim that VP brings embeddings closer, and the caption contains a broken cross-reference.

Figure 2

Figure 2 effectively communicates the scaling benefits of visual pretraining across four distinct metrics. However, the caption's claim of 'comparable CPT loss' in panel (a) is not visually supported by the plot's layout, and the x-axis break in panel (b) is not clearly indicated in the tick labels.

Figure 3

The figure displays clear data trends regarding visual token budget and training resolution, but the caption is severely flawed with raw LaTeX artifacts, repetition, and truncated text that obscures the intended explanation.

Figure 4

The figure illustrates attention maps for text and visual reasoning but suffers from a mismatch between the caption's description of the source tokens and the highlighted regions in the top panel. Additionally, the visual overlay in the bottom panel lacks sufficient clarity to verify token-level alignment.

Figure 5

The figure effectively contrasts the two architectural pathways, but the legend is disconnected from the diagram elements, and the input source images are too blurry to be legible.

Action items.

- [**science**] Figure 1: The caption states that VP brings matched visual and textual document embeddings closer, but the top t-SNE plot (representing the TP pathway) shows the clusters already overlapping significantly, while the bottom plot (VP) shows them more separated. This contradicts the caption's claim that VP brings them closer.

- [**writing**] Figure 1: The caption contains a broken cross-reference ('quantified in : VP') where a figure or section number is missing.
- [**writing**] Figure 2: The caption for panel (a) claims 'comparable CPT loss' to justify the comparison, but the plot shows two distinct loss curves (VP vs. TP) that are not aligned or normalized to a common loss baseline, making the 'favorable SFT trajectory' claim difficult to verify visually without further context.
- [**writing**] Figure 2: Panel (b) x-axis contains a break symbol ('//') between 20 and 76, but the axis tick labels are not adjusted to reflect the non-linear spacing, which can mislead the reader about the density of data points in that region.
- [**writing**] Figure 2: Panel (b) includes specific annotations (e.g., 'AIME-25: 2.88x') directly on the plot area without a clear legend entry or pointer line connecting them to the specific data series, relying on color matching which may be ambiguous.
- [**writing**] Figure 3: The caption contains raw LaTeX formatting artifacts ('1$$') and appears to be a truncated draft with repeated sentences and cut-off text at the end.
- [**science**] Figure 3: The left panel's x-axis label 'Visual token budget (x production)' is ambiguous and likely a typo for 'x production budget' or similar, failing to clearly define the scaling factor relative to a baseline.
- [**writing**] Figure 3: The caption text is repetitive and contains incomplete sentences (e.g., 'inflate hard-negative saturation in the'), indicating a copy-paste error or unfinished editing.
- [**writing**] Figure 4: The caption states attention is measured 'from the final answer sentence to previous tokens,' but the top panel shows a red bounding box around the question constraint ('Find the remainder...') rather than the answer sentence, creating a mismatch between the described mechanism and the visual evidence.
- [**writing**] Figure 4: The bottom panel's attention map is overlaid on the text with low contrast and no clear grid alignment, making it difficult to verify which specific visual tokens correspond to the highlighted semantic regions.
- [**writing**] Figure 5: The legend at the bottom right defines 'Generated', 'Current', and 'Future' tokens, but the diagram does not explicitly label which specific blocks correspond to these categories, forcing the reader to guess the mapping based on color alone.
- [**writing**] Figure 5: The input images on the far left and right are extremely low-resolution and illegible, making it impossible to verify the 'visual pretraining' context described in the caption.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The paper is generally well-structured for a technical audience, but it relies on several acronyms and specific notation that are introduced without explicit definition at their first point of use, creating minor barriers for a competent reader from an adjacent field (e.g., a researcher in NLP reading a CV paper or vice versa).

Specifically, the acronyms **VP** (Visual Pretraining) and **TP** (Text Pretraining) are used frequently in the Introduction and Results sections before being formally defined. While the full terms appear in the abstract, the acronyms themselves are not expanded until later in the text. A reader skimming the Introduction would encounter "VP" and "TP" without immediate context.

Similarly, **CPT** (Continued Pretraining) and **SFT** (Supervised Fine-Tuning) are used in Section 2 without expansion. While these are standard in the broader LLM community, they are not universal across all adjacent disciplines (e.g., traditional computer vision or statistics).

The term **InfoNCE** is used in the Methods and Results without a brief definition or expansion (e.g., "InfoNCE loss"). While a standard contrastive objective, the acronym itself is not self-explanatory to all researchers.

Finally, the notation **Linear CKA** and **Mutual k-NN** appears in the Results section. While "k-NN" is widely known, "CKA" (Centered Kernel Alignment) is a specific metric that should be defined upon first introduction to ensure clarity. The symbol τ is used for temperature in the loss function equations but is not explicitly defined in the text immediately preceding or following the equations, which could cause momentary confusion.

These issues are easily resolved by expanding acronyms at first use and adding brief parenthetical definitions for specific metrics and symbols, ensuring the paper is self-contained for the target audience.

Action items.

- [**writing**] Section 1 (Introduction) and Section 2 (Results) use the acronym 'VP' and 'TP' repeatedly before they are explicitly defined as 'Visual Pretraining' and 'Text Pretraining'. While the abstract mentions 'Visual Pretraining', the acronyms themselves are not introduced until Section 2. Define them at first use in the Introduction (e.g., 'Visual Pretraining (VP)') to ensure immediate clarity for adjacent-field readers.

- [**writing**] Section 2 (Results) introduces 'CPT' (Continued Pretraining) and 'SFT' (Supervised Fine-Tuning) without expansion. While common in the subfield, a competent adjacent-field PhD (e.g., from NLP or Computer Vision) might not instantly map these specific acronyms to the full terms in this context. Expand them at first occurrence: 'continued pretraining (CPT)' and 'supervised fine-tuning (SFT)'.

- [**writing**] Section 2 (Results) and Section 4 (Methods) use 'InfoNCE' without defining it. While a standard contrastive loss, the specific acronym is not universally known outside contrastive learning subfields. Add a brief gloss at first use: 'InfoNCE (a contrastive loss function)' or 'InfoNCE (Noise-Contrastive Estimation)'.

- [**writing**] Section 2 (Results) mentions 'Linear CKA' and 'Mutual k-NN' without defining the acronyms. 'CKA' (Centered Kernel Alignment) and 'k-NN' (k-Nearest Neighbors) are standard but specific. Define them at first use: 'Linear Centered Kernel Alignment (CKA)' and 'Mutual k-Nearest Neighbor (k-NN) overlap'.

- **[writing]** Section 4 (Methods) and Appendix A1 use the symbol ***au*** for temperature in the InfoNCE loss (Eq. 4, Eq. A11) without explicitly stating 'where ***au*** is the temperature parameter' in the immediate text surrounding the equation. While standard, explicitly defining it near the equation prevents ambiguity for readers from adjacent fields.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a coherent argument that visual pretraining (VP) on raw scientific documents improves reasoning compared to text-only pretraining (TP) on extracted text, supported by a matched-corpus experimental design. The logical flow from the problem (information loss in text extraction) to the solution (VP) and evidence (benchmark gains) is generally sound.

However, there is a logical inconsistency regarding the "efficiency" claim in the Results section versus the Methods section. In Section 2, the authors state: "VP surpasses TP while consuming only 25% of the token budget." This phrasing implies the *total* training token budget for VP was 25% of TP. Yet, Section 4 ("Training setup") clarifies that total CPT budgets were 180B tokens for TP and 120B for VP. The 25% figure (20B visual vs 80B text) applies *only* to the scientific-PDF subset, not the entire corpus.

This creates a gap where the conclusion in Section 2 (total budget efficiency) does not strictly follow from the premises in Section 4 (subset efficiency). The argument overgeneralizes the efficiency gain to the total training cost, which is not supported by the numbers. The text must clarify that the 25% figure refers specifically to the scientific-PDF representation efficiency, not the aggregate training budget, to ensure the conclusion logically follows from the stated premises.

Action items.

- **[writing]** Section 2 claims VP uses 'only 25% of the token budget' compared to TP. However, Section 4 clarifies total CPT budgets are 180B (TP) vs 120B (VP). The 25% figure applies only to the scientific-PDF subset (20B vs 80B), not the total budget. The Results phrasing misleadingly implies a total budget reduction. Clarify that the efficiency gain applies specifically to the PDF corpus representation, not the aggregate token count.

- **[writing]** Section 2 states VP trains on 20B visual tokens vs TP's 80B text tokens to support efficiency. But Section 4 shows total CPT budgets are 120B (VP) vs 180B (TP). The argument conflates subset efficiency with total training scale, implying a 75% cost reduction when it was only ~33%. Distinguish between the efficiency of the PDF representation and the total training budget to avoid logical overreach.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper generally maintains a strong link between its evidence and its core findings regarding scientific document pretraining. However, the rhetoric in the Abstract, Introduction, and Discussion occasionally widens the scope of the claims beyond the specific experimental conditions (scientific PDFs, specific benchmarks, and model families).

In the **Abstract**, the phrase “scalable learner for foundation model intelligence” implies a universal capability across all domains of intelligence, whereas the evidence is strictly confined to scientific reasoning benchmarks derived from PDF documents. Similarly, stating that the method “consistently outperforms text-only pretraining” without qualification suggests a universal superiority, while the results show gains primarily on structure-heavy scientific tasks and marginal or mixed results on others (e.g., HLE).

The **Introduction** claims to establish VP as a pathway for “learning both language and visual intelligence.” While the paper does show improvements on multimodal benchmarks, these are limited to specific datasets (MMMU-Pro, ChartQAPro) and two model families. The language suggests a broader paradigm shift in visual intelligence than the specific benchmark results license.

Most notably, the **Discussion** section speculates that “foundation models could be trained primarily on large-scale visual streams” based on the efficiency observed with scientific PDFs. This extrapolates a trend from a highly structured, text-dense domain to general visual corpora (like natural images or video), which the paper explicitly acknowledges as an open question in its limitations but then proceeds to frame as a likely future paradigm in the outlook. This creates a tension between the stated limitations and the forward-looking rhetoric.

These issues are primarily matters of **writing** (scope and hedging) rather than fatal flaws in the science, as the core results are sound within their tested boundaries. Narrowing the claims to reflect the specific domain (scientific documents) and the specific nature of the “visual intelligence” gains (benchmark-specific) would align the rhetoric with the evidence.

Action items.

- [**writing**] Abstract: The claim that visual pretraining is a ‘scalable learner for foundation model intelligence’ and ‘consistently outperforms text-only pretraining’ is too broad. Experiments are limited to scientific PDFs and specific benchmarks (GPQA, MMLU-Pro, AIME, HLE). Replace ‘foundation model intelligence’ with ‘scientific reasoning’ and qualify ‘consistently’ to ‘on scientific reasoning benchmarks under matched corpora’.
- [**writing**] Introduction: The statement that VP ‘establishes VP as a scalable pathway

for learning both language and visual intelligence' overgeneralizes the results. The visual intelligence gains are demonstrated only on specific multimodal benchmarks (MMMU-Pro, ChartQAPro) using two model families (Qwen, Llama). Narrow the claim to 'demonstrates a pathway for improving scientific reasoning and specific multimodal tasks'.

- [**writing**] Discussion: The outlook suggests 'foundation models could be trained primarily on large-scale visual streams' based on results from a single domain (scientific PDFs). This extrapolation to general 'large-scale visual streams' (e.g., natural images, video) is not supported by the data. Add a limitation explicitly stating that the scalability to non-document visual corpora remains untested.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

This paper presents a method for visual pretraining (VP) on raw scientific document pages to improve language reasoning and multimodal alignment. The methodology involves rendering PDFs as images, extracting foreground visual tokens via a frozen vision encoder, and training an LLM to predict the next visual latent. The data sources are described as publicly accessible PDFs indexed by Common Crawl and standard open-source text corpora (FineWeb-Edu).

From a safety and ethics perspective, the work does not present foreseeable, non-trivial risks of harm that are unaddressed. The research focuses on improving scientific reasoning and document understanding, which are generally benign capabilities. The method does not involve:

1. **Human subjects or PII:** The data consists of scientific documents (papers, textbooks) which are public records. The paper does not mention collecting private data, conducting surveys, or using personally identifiable information (PII) from individuals. The "100 image-text pairs" used for alignment analysis are drawn from a held-out set of scientific PDFs, not private user data.

2. **Dual-use for harm:** The capability to reason better about scientific diagrams and equations does not lower the barrier to creating biological weapons, cyberattacks, or disinformation in a specific, actionable way that differs from general LLM capabilities. The paper does not describe generating harmful content or exploiting vulnerabilities.

3. **Deception or Surveillance:** The system is not designed to impersonate humans, manipulate user behavior covertly, or surveil individuals.

4. **License Violations:** The paper states the data comes from public sources (Common Crawl, FineWeb-Edu) and uses standard parsing tools (MinerU2.5). There is no indication of scraping data in violation of Terms of Service or redistributing copyrighted content in a way that the paper fails to disclose.

The paper includes a "Limitations" section (Section 3) discussing the scope of the method (e.g., focus on scientific documents, not natural images) and future work, which is appropriate. No specific safety disclosures, IRB statements, or mitigation strategies are required for this type of low-risk, algorithmic research on public scientific data. The verdict is `accept` with no action items.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a compelling hypothesis that visual pretraining (VP) on raw scientific documents improves reasoning capabilities compared to text-only pretraining (TP) on the same corpus. The experimental design generally follows a matched-corpus approach, which is a strong starting point. However, several critical gaps in the experimental design prevent the evidence from fully supporting the magnitude and specificity of the claims made.

First, the statistical robustness of the reported improvements is unclear. Table 1 (tab:main_result_text) presents specific point estimates for gains (e.g., +3.22 on GPQA, +2.1 on MMLU-Pro) without reporting standard deviations, confidence intervals, or the number of random seeds used for the non-AIME benchmarks. While the text mentions "32 runs" for AIME-25, it is silent on the variance for the other benchmarks. In large-scale LLM evaluations, performance can fluctuate significantly based on random initialization, sampling temperature, or prompt variations. Without reporting variance across multiple seeds, it is impossible to determine if the observed gains are statistically significant or within the noise floor of the evaluation protocol. The authors must report mean ± standard deviation (or standard error) across at least 3-5 independent training runs for all benchmarks to establish that the effect is stable and not a result of lucky seeds.

Second, the claim that the improvement is due to the *visual representation* rather than the *total training budget* is confounded. The paper explicitly states that VP uses 20B visual tokens while TP uses 80B text tokens from the same PDFs, resulting in a total CPT budget of 120B for VP versus 180B for TP. The authors argue this difference reflects the "compactness" of visual tokens, but they do not control for the total amount of optimization steps or data volume. It is equally plausible that the performance gain arises because the VP model was trained on a different (potentially more optimal) total token count, or that the specific ratio of visual-to-text tokens in the VP mix (1:4) is simply a better training recipe than the TP mix, rather than the visual modality itself being superior. To isolate the contribution of the visual modality, the authors should include a control experiment where the TP baseline is trained on a matched token budget (e.g., 120B tokens) or where the VP model is trained on a matched 180B budget. Without this, the "efficiency" claim is ambiguous.

Third, the attribution of the gain to the specific VP mechanism (next-visual-latent prediction) lacks ablation. The paper introduces a complex pipeline involving foreground filtering, raster ordering, and a contrastive loss. However, there is no ablation study that removes or varies these components in isolation. For instance, does the gain persist if the visual tokens are not filtered for foreground? Does the contrastive loss matter compared to a simple MSE reconstruction? Without these controls, the claim that the specific "visual pretraining" paradigm is the driver of the improvement is not fully supported; the gain could be an artifact of the specific hyperparameters or the inclusion of any visual signal, regardless of the specific objective.

Finally, the evaluation of "visual structure density" (Figure 2c) relies on a post-hoc categorization of the test set. While the trend is consistent with the hypothesis, the paper does not provide a negative control or a randomized baseline to ensure that the density metric itself isn't correlated with other confounding factors (e.g., question difficulty or length).

In summary, while the direction of the results is promising, the evidence is currently insufficient to rule out alternative explanations such as training budget differences, random seed variance, or the specific tuning of the visual pipeline. Addressing these design gaps is necessary to substantiate the central claims.

Action items.

- [**writing**] The paper presents a compelling hypothesis that visual pretraining (VP) on raw scientific documents improves reasoning capabilities compared to text-only pretraining (TP) on the same corpus. The experimental design generally follows a matched-corpus approach, which is a strong starting point. However, several critical gaps in the experimental design prevent the evidence from fully supporting the magnitude and specificity of the claims made. First, the statistical robustness of the reported impro

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical reporting in this paper is currently insufficient to support the magnitude of the claims made regarding the superiority of Visual Pretraining (VP) over Text Pretraining (TP). While the experimental design (matched corpora) is sound, the **statistical treatment of the results** lacks necessary uncertainty quantification and inferential rigor.

1. Missing Uncertainty Estimates In Section 2 ("Results") and Tables 1 & 2, the authors report point estimates for benchmark scores (e.g., "79.29" on GPQA, "90.21" on AIME-25). The text mentions that AIME-25 scores are "averaged over 32 runs," yet **no standard deviation (SD), standard error (SE), or confidence intervals (CI)** are reported for these averages. Similarly, GPQA is reported as "average pass@8," but the variance across seeds or runs is omitted.

- **Impact:** Without error bars, it is impossible to determine if the reported improvements (e.g.,

+3.22 points on GPQA) are statistically significant or within the noise of the evaluation process. A difference of 3 points could be trivial if the SD is 5, or highly significant if the SD is 0.5.

- **Fix:** Report all benchmark results as `Mean ± SD` (or `Mean ± SE`) in the tables. If the 32 runs for AIME-25 were performed, the SD must be included.

2. Lack of Statistical Significance Testing The paper repeatedly uses language implying statistical significance (e.g., "consistently outperforms," "substantially more efficient") without providing the corresponding statistical tests.

- **Impact:** The claim that VP is "significantly better" is unsupported by the data presented. With 4 backbones and 4 benchmarks, there are 16 pairwise comparisons. Highlighting the "best" improvements without correcting for multiple comparisons inflates the false-positive rate.
- **Fix:** Perform paired statistical tests (e.g., paired t-test or Wilcoxon signed-rank test, depending on normality) comparing VP vs. TP for each backbone/benchmark pair. Report the p-values. Given the number of comparisons, apply a correction method (e.g., Holm-Bonferroni or Benjamini-Hochberg) and indicate which results remain significant. Alternatively, if the authors choose not to run formal tests, they must soften the language to "observed improvements" and avoid terms like "significantly" or "substantially" unless qualified by the magnitude of the effect size relative to the reported variance.

3. Token Budget Comparison Validity The paper claims VP is "more efficient" because it achieves better results with 20B tokens vs. 80B for TP. However, the statistical comparison of performance across *different* token budgets is confounded.

- **Impact:** It is unclear if the gain is due to the *visual modality* or simply the fact that the visual representation is more information-dense. A fairer statistical comparison would involve running TP at the 20B token budget to see if the modality itself drives the gain, or normalizing performance by token count.
- **Fix:** While this borders on experimental design, the statistical claim of "efficiency" is currently based on a single data point comparison (20B vs 80B). The authors should either include an ablation where TP is run at 20B tokens to isolate the modality effect, or rephrase the efficiency claim to be strictly descriptive of the specific run conditions without implying a generalizable statistical superiority of the modality at that specific budget.

In summary, the paper's quantitative claims are currently "point estimates without error bars." Adding uncertainty metrics and significance tests is a necessary step to validate the reported gains.

Action items.

- [**writing**] Section 2 reports 'average pass@8 for GPQA' and 'average score over 32 runs for AIME-25' but provides no standard deviation, standard error, or confidence intervals for these means. Without uncertainty estimates, the reported point differences (e.g., +3.22 on

GPQA) cannot be judged for stability or statistical significance. Report mean ± SD (or 95% CI) for all benchmark scores in Table 1 and Table 2.

- [**writing**] Table 1 and Table 2 present single-point improvements (e.g., '76.24 to 79.29') without indicating whether these differences are statistically significant. Given the multiple comparisons across 4 backbones and 4 benchmarks (16 pairwise tests), the paper should either report p-values from appropriate paired tests (e.g., paired t-test or Wilcoxon signed-rank) with multiple-comparison correction (e.g., Holm-Bonferroni), or explicitly state that the results are descriptive without inferential claims.
- [**science**] The claim that VP uses 'only 25% of the token budget' (Section 2) compares 20B visual tokens to 80B text tokens. However, the statistical validity of comparing performance across different token budgets is questionable without a scaling analysis or normalization. The paper should clarify if the 25% figure is a strict constraint or an observation, and whether the performance gains hold when TP is also run at the 20B token budget (an ablation missing from the reported tables).

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The paper is generally well-written, with clear and concise prose. However, there are a few areas where the flow could be improved. In Section 2, the second paragraph contains a fragmented sentence that should be merged for better readability. Additionally, several sentences in Sections 2, 3, and 4 are long and list-heavy, which can interrupt the flow. These sentences could be split or rephrased for clarity. For example, in Section 2, the sentence listing the metrics for different benchmarks is a bit long and could be split into two. Similarly, in Section 3, the sentence describing the limitations of the study is a bit wordy and could be tightened. In Section 4, the description of the method is clear, but some sentences are a bit vague and could be clarified. Overall, the paper is well-written, but a few minor revisions would improve its readability.

Action items.

- [**writing**] The paper is generally well-written, with clear and concise prose. However, there are a few areas where the flow could be improved. In Section 2, the second paragraph contains a fragmented sentence that should be merged for better readability. Additionally, several sentences in Sections 2, 3, and 4 are long and list-heavy, which can interrupt the flow. These sentences could be split or rephrased for clarity. For example, in Section 2, the sentence listing the metrics for different benchmarks is